

Muhsina Tarannum Munfa

PhD Applicant | AI/ML Researcher,

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RESEARCH INTEREST

My research interests lie in low-resource and multilingual natural language processing, AI for healthcare, and explainable machine learning. I am particularly interested in developing robust, human-centered AI systems for underrepresented languages and real-world applications.

EDUCATION

Metropolitan University

BSc in Computer Science & Engineering

GPA: 3.56/4.00

2018 - 2022

Sylhet, Bangladesh

PUBLICATIONS& PREPRINTS

A Dual Pipeline Machine Learning Framework for Automated Multi Class Sleep Disorder Screening Using Hybrid Resampling and Ensemble Learning || Md Sultanul Islam Ovi, **Muhsina Tarannum Munfa**, G.M.M Miftahul Alam Adib, Syed Sabbir Hasan (2026) || (Arxiv) 📄

Accepted: Social Network Analysis and Mining

- Implemented and evaluated machine learning models for automated multi-class sleep disorder prediction
- Developed data preprocessing, model training, and performance evaluation pipelines using Python and scikit-learn
- Contributed to the study design and early manuscript development, including drafting the initial methodology

Explainable Machine Learning for Chronic Kidney Disease Classification from Routine Urinalysis and Diabetes Records || **Muhsina Tarannum Munfa**, G.M.M Miftahul Alam Adib, Md Sultanul Islam Ovi, (2026) || (Research Square) 📄

- Developed explainable ML models for CKD classification using two real-world medical datasets.
- Evaluated model performance, calibration, robustness, and prediction stability across clinical information sets.
- Applied SHAP explanations to identify influential predictors and assess explanation stability across model runs.

ONGOING RESEARCH PROJECTS

Bangla Mental Health NLP and Synthetic Dataset Development

Research Collaborator

- Developing an ontology-driven approach to generate synthetic Bangla, Banglish, and English mental-health text.
- Investigating label consistency, dataset quality, and evaluation methods for low-resource mental-health NLP.

Energy-Aware Small Language Models for Evidence-Grounded Climate Misinformation Detection

Research Collaborator

- Collaborating on research exploring AI and data-driven methods for climate and sustainability challenges.
- Contributing to research planning, literature review, and identification of relevant datasets.

FinReason: Evaluating Causal, Temporal, Multi-Hop, and Counterfactual Reasoning in Financial Large Language Models

Research Lead

- Leading research planning and coordination for projects at the intersection of AI and financial applications.
- Exploring research problems involving machine learning, evaluation, and responsible AI in financial contexts.

RESEARCH EXPERIENCE

ResearchDen 📄 | Research Collaborator

Lead Researcher & Supervisor: **Md. Sultanul Islam Ovi**

2025 - Present

Remote

- Contributed to the initial research design and manuscript preparation for multi-class sleep disorder prediction
- Contributing to dataset creation, preprocessing, and experimental design for predictive modeling studies.
- Assisting in the development and evaluation of explainable AI (XAI) methods for healthcare applications.

Prediction of Customers' Purchase Intention in E-commerce Stores

B.Sc. Thesis | Metropolitan University | Supervisor: Akhlak Uz Zaman Ashik, Assistant Professor

Jan 2022 – Aug 2022

Sylhet, Bangladesh

- Investigated customer purchase intention prediction using machine learning on an e-commerce dataset
- Implemented and compared Logistic Regression, KNN, SVM, Naive Bayes, Decision Tree, Random Forest, and ANN models.
- Identified Random Forest and ANN as the best-performing models.

WORK EXPERIENCE

FlyRank AI

Backend AI Engineering Intern

Jul 2026 – Present
Remote

- Collaborating with the engineering team on AI system implementation and backend development.
- Developing backend components for AI-powered applications using Python and modern API frameworks.

YES Associate

Admission Support Officer

2023 - 2024
Sylhet, Bangladesh

- Prepared and reviewed application documents for students applying to international universities.
- Assisted applicants throughout the admissions documentation and application process.

OpenGenus IQ

Research Scientist, Intern

2022 - 2023
Remote

- Conducted literature reviews on computer science and machine learning topics using technical and educational resources.
- Produced technical articles by synthesizing information from multiple sources into clear explanations.

PROJECTS

Curriculum-Based Bangla QA System | Project Lead | Live Demo

ChromaDB | FastAPI | Gemini-3.5-flash-lite

- Leading the development of an NLP system for generating Bangladeshi curriculum-grounded exam questions
- Coordinating a team-based implementation using Python, ChromaDB, LLMs, and Bangla NLP models
- Designing the retrieval, question-generation, and answer-validation pipeline to improve factual grounding and curriculum alignment

Air Quality & Health Risk Assessment System | Live Demo

FastAPI | PostgreSQL | XGBoost | Streamlit | Looker Studio

- Built an end-to-end system for AQI prediction, integrating external APIs, XGBoost, PostgreSQL, and FastAPI
- Configured PostgreSQL on Neon with schema for time-series AQI storage and managed deployment on Render
- Developed interactive Streamlit and Looker Studio dashboards for real-time air quality monitoring and health risk visualization.

ChefBot | Live Demo

LangChain | ChromaDB | Sentence Transformers | Groq API | Gradio

- Developed a RAG-based cooking assistant using LangChain, ChromaDB, and LLaMA 3.
- Implemented document ingestion, embedding generation, and retrieval pipelines for cookbook-based question answering.
- Reduced hallucinations through prompt engineering and context-constrained response generation.

TECHNICAL WRITING

ResearchDen Blog

Authored four technical articles on

- The Writing Mistakes in Almost Every First Paper
- Imposter Syndrome in a Room Full of Smart People
- Building Your First Dataset, and the Responsibility That Comes With It
- Working With Low-Resource and Multilingual Text

SKILLS

- **Programming Languages:** Python, C++, JavaScript
- **Machine Learning & Deep Learning:** Scikit-learn, HuggingFace Transformers PyTorch
- **NLP & LLM:** LangChain, ChromaDB, FAISS, Groq API
- **Backend & Data Engineering:** FastAPI, PostgreSQL, MongoDB, REST APIs, SQL
- **Deployment & MLOps:** Streamlit, Gradio, Hugging Face Spaces, Render
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn,
- **Tools & Platforms:** Git, GitHub, Google Colab, VS Code, Looker Studio, LaTeX, Typst

ACADEMIC SERVICES

Conference Reviewer | 2026

- NGEN-AI 2026, Italy
- QUANCOM 2026, Italy

Peer Reviewer | 2026

- European Journal of Computer Sciences and Informatics

CERTIFICATIONS

- **Research Project Lifecycle Workshop** | Organized by AI/ML Professional Community Bangladesh
- **Research Scientist, Intern at OpenGenus**

October - 2025
April - 2023